

JIAYI ZHAO

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Education

University of Washington, Seattle

Sep. 2025 -

Ph. D of Electrical Engineering

GPA: 3.92

University of Michigan, Ann Arbor

Sep. 2023 - May 2025

Bachelor of Science and Engineering in Data Science & Engineering (Mathematics Minor)

GPA: 3.72/4.0

University of Michigan Shanghai Jiaotong University

Sep. 2021 - August 2025

Bachelor of Engineering in Electrical and Computer Engineering

GPA: 3.28/4.0

Course Modules: Robot Control, Control Theory for Biological Sensorimotor Systems, Numerical Methods, Natural Language Processing, Computer Vision, Entropy and Information, Data Structures and Algorithms, Introduction to Computer Organization, Applications of Stochastic Processes

Interests

- Autonomous systems
- Multi-agent systems
- Machine learning
- Distributed Controls
- Biomechanical Sensorimotor Control

Publications

Human Balancing on a Log: A Switched Multi-Layer Controller

Jiayi Zhao, Mo Yang, Jing Shuang (Lisa) Li

Area: Bio-mechanical control, nonlinear control systems, switched and hybrid systems.

Accepted & Presented ACC 2025.

Projects

Switched multi-Layer controller for human postural balancing on uneven terrain

Feb. 2024 - Oct. 2024

- Designed a three-case multi-layer controller with the inspiration from the human motor-sensory feedback control loop
- Developed a fully physically realistic model; designed and simulated the model using MATLAB
- Verified the Controller in MATLAB and tested noise-resistance

Large Able-bodied Dataset Standardization

May 2024 - Oct. 2024

- Designed python pipeline for processing multiple datasets; integrated multiple non-parquet formatted datasets into a complete compatible parquet file
- Assisted in setting the standard of the data
- Verified the data with OpenSim simulation

Micro Spiking Neural Network Controller

Sep. 2023 - May 2024

- Designed and implemented a RL-based controller for single leg under SNN architecture
- Designed and trained the model with python and torch
- Simulated the model in python

Passive Exoskeleton CTM Tennis Elbow Rehabilitation Device

Sep. 2023 - May 2024

- Designed and optimized the Constrained Torque Model (CTM) based on detailed force and physics analysis of the Polycarbonate (PC) material in elastic deformation
- Designed the overall device designed with Solidworks
- Manufactured the model with 3D printing and performed effectiveness and comfort testing on participants

Experience

Chairman Consultant <i>SJTU Student Science, Technology and Innovation Association</i> - Planned and launched the freshman’s AI contest - Organized and coordinated the freshman’s Robotic competition	Sep. 2022 - Aug. 2023
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Honors and Awards

University Honors <i>University of Michigan Dean’s list. Recognizes exceptional academic achievement across the entire university.</i>	2024
Dean’s List <i>University of Michigan Dean’s list. Recognizes high academic performance within specific college or school.</i>	2024
University Honors <i>University of Michigan Dean’s list. Recognizes exceptional academic achievement across the entire university.</i>	2023
Dean’s List <i>University of Michigan Dean’s list. Recognizes high academic performance within specific college or school.</i>	2023
Honorable Mention in MCM <i>A mathematical modelling contest for college students around the globe. Honor for top 21% participants</i>	2023
Finalist in IMMC (international) <i>A mathematical modelling contest for high school students around the globe. Honor for top 1.5% participants.</i>	2021
Meritorious Winner in HiMCM <i>A mathematical modelling contest for high school students around the globe. Honor for top 8% participants</i>	2019

Technical Skills

Languages: Python, Matlab, C++, C#
Developer Tools: VS Code, Git, OpenSim, Unity, SQL, $\LaTeX$